3-11-26 (WEEK 3)

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DISCRIMINANT AND INTEGRAL BASES

REMARK

For a number field $K/\mathbb{Q}$ and $\mathcal{O}_K$: free $\mathbb{Z}$-mod of $rk = [K:\mathbb{Q}]$

⚠ $\mathbb{Q} \subseteq K \subseteq L$ are ext of number fields

So $\mathcal{O}_K, \mathcal{O}_L$: free $\mathbb{Z}$-mod, but $\mathcal{O}_L$ is NOT necessarily a free $\mathcal{O}_K$-mod ($\mathcal{O}_K$ may NOT be a PID)

<u>How to find the $\{e_i\}$ basis</u>

For $K/\mathbb{Q}$ with $e_i \in \mathcal{O}_K$, then consider Tr form on $\bigoplus_i \mathbb{Z}e_i$

If we know the index $[L:\mathcal{O}_K] = d < \infty$, where $\mathcal{O}_K \subseteq \bigoplus \mathbb{Z}\frac{e_i}{\det M}$ (M: Matrix of Tr form on $\bigoplus \mathbb{Z}e_i$) L

Then, for $[K:\mathbb{Q}] = n$, $K = \mathbb{Q}(\theta)$, $\theta \in K$, $\alpha = \sum_{i=0}^{n-1} c_i \theta^i$, how do we find min poly of α over $\mathbb{Q}$ with min poly of θ?

DEFINITION

(L, b) is a $\mathbb{Z}$-lattice if L is a free $\mathbb{Z}$-mod of finite rk and <u>b: sym bilinear nondegenerate form</u>, s.t. $L \times L \xrightarrow{b} \mathbb{Z}$ (has a basis)

In particular, $(\mathcal{O}_K, Tr|_{\mathcal{O}_K})$ is a lattice

PROPOSITION

Let (L, b): lattice, where $\{e_i\}_{i=1}^n$ and $\{\varepsilon_i\}_{i=1}^n$ are bases of L over $\mathbb{Z}$

Denote M_e, M_ε: matrix rep of b w.r.t. bases $\{e_i\}$ and $\{\varepsilon_i\}$ respectively.

Then, $\det(M_\varepsilon) = \det(M_e)$

<u>Proof</u>

Notice, $\varepsilon_i = \sum_r a_{r,i} e_r$

Thus, $(M_\varepsilon)_{ij} = b(\varepsilon_i, \varepsilon_j) = [a_{1i}, \dots, a_{ni}]\, M_e \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix} = (A^T M_e A)_{ij}$. Notice, A: invertible because the bases are both interchangable. (col vec)

We get $\det(M_\varepsilon) = \det(A)^2 \det(M_e) = \det(M_e)$ since $A \in GL_n(\mathbb{Z}) \Rightarrow \det(A) = \pm 1$ □ (invertible matrix; e.g. $\det(A) = 2$, Then $\det(A^{-1}) = \frac{1}{2} \notin \mathbb{Z}$)

DEFINITION

(i) We define the discriminant of $K/\mathbb{Q}$ to be the <u>determinant of Tr form</u> on $\mathcal{O}_K$ (In some applications, discriminant is $(-1)^{\frac{n(n-1)}{2}} \cdot \det$, $n = [K:\mathbb{Q}]$)

We denote this by $disc(K/\mathbb{Q})$ or $disc(\mathcal{O}_K)$

(ii) More generally, $\forall$ lattice (L, b), we can define $disc(L)$ in a similar way

EXAMPLE

$L := (\mathbb{Z} \oplus \mathbb{Z}\sqrt{d}, Tr)$, d: squarefree int

Basis $= \{1, \sqrt{d}\}$

Notice, $Tr(1,1) = Tr_{K/\mathbb{Q}}(1) = tr\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ ($1(1)$, $\sqrt{d}(1)$)

By similar logic, $disc(L) = \det\begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix} = 4d$

EXAMPLE

$\mathbb{Z} \oplus \mathbb{Z}\alpha$, $\alpha^2 + a\alpha + b = 0$, $a, b \in \mathbb{Z}$ and $a^2 - 4b \notin \mathbb{Q}^2$

(L, Tr) disc = ?

Basis $= \{1, \alpha\}$

$Tr(1,1)$: $1(1 \cdot 1) = 1$, $\alpha(1 \cdot 1) = \alpha \Rightarrow Tr(1,1) = tr\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 2$

$Tr(1,\alpha)$: $1(1 \cdot \alpha) = \alpha$, $\alpha(1 \cdot \alpha) = \alpha^2 = -a\alpha - b \Rightarrow Tr(1,\alpha) = tr\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix} = -a$

$Tr(\alpha,1)$: $Tr(\alpha,1) = -a$ like above

$Tr(\alpha,\alpha)$: $1(\alpha \cdot \alpha) = \alpha^2 = -a\alpha - b$, $\alpha(\alpha \cdot \alpha) = -a\alpha^2 - b\alpha = -a(-a\alpha - b) - b\alpha = (a^2 - b)\alpha + ab \Rightarrow Tr(\alpha,\alpha) = tr\begin{pmatrix} -b & ab \\ -a & a^2-b \end{pmatrix} = a^2 - 2b$

$\therefore disc(L) = \det\begin{pmatrix} 2 & -a \\ -a & a^2-2b \end{pmatrix} = a^2 - 4b$

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EXAMPLE

In general, what is disc for $(\mathbb{Z}(\alpha), Tr)$, where $x^n+ax+b=0$, $a,b\in\mathbb{Z}$?

For $K=\mathbb{Q}(\theta)$, we get the Vandermonde matrix
$\operatorname{disc}(Tr(-,-)) = (-1)^{\frac{n(n-1)}{2}} \prod_{i\neq j} (\sigma_i(\theta)-\sigma_j(\theta))$, σ_i: all $\mathbb{Q}$-embedding of $K \hookrightarrow L$ (L: Galois closure of K)

Assume θ is integral with $f_{\theta,\mathbb{Q}}(x) = \sum_{i=0}^{n} a_i x^i = \prod_i (x-\sigma_i(\theta))$
Then, $\operatorname{disc}(Tr(-,-)) = (-1)^{\frac{n(n-1)}{2}} f'_{\theta,\mathbb{Q}}(\sigma_i(\theta))$

PROPOSITION

Let (L,b): lattice, $L'\subseteq L$ be a sublattice $(L', b|_{L'})$
Assume $[L:L']<\infty$, i.e. $\operatorname{rk}_{\mathbb{Z}} L = \operatorname{rk}_{\mathbb{Z}} L'$
Then, $\operatorname{disc}(L') = \operatorname{disc}(L) = [L:L']^2$

<u>Proof</u>

By <u>structure thm</u>, $\exists$ basis of L $\{e_i\}$ s.t. $L' = \bigoplus_i \mathbb{Z} b_i e_i \subseteq L = \bigoplus_i \mathbb{Z} e_i$, $b_i \in \mathbb{Z}_{>0} \Rightarrow [L:L'] = \prod_{i=1}^{n} b_i$

module over PID

For $b|_{L'}$, M': matrix rep w.r.t. $\{b_i e_i\}$
Then, $M'_{ij} = b(b_i e_i, b_j e_j) = b_i b_j\, b(e_i, e_j) = b_i b_j M_{ij}$, where M: matrix rep of b w.r.t. $\{e_i\}$.

$$\Rightarrow M' = \begin{pmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{pmatrix} M \begin{pmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{pmatrix}$$

$$\Rightarrow \det(M') = \det(M) \underbrace{[L:L']^2}_{\prod_{i=1}^{n} b_i} \quad \square$$

EXAMPLE

Say $K=\mathbb{Q}(\theta)$, $\theta^3-\theta-1=0$
$\mathcal{O}_K$ basis?

Notice, $\underbrace{\mathbb{Z}[\theta]}_{\operatorname{rk} 3} \subseteq \mathcal{O}_K \Rightarrow [\mathcal{O}_K : \mathbb{Z}[\theta]] < +\infty$

means $\nexists$ sublattice that extended to it, otherwise $[L':L]^2$ being a factor will mean it's not squarefree

As $\operatorname{disc}(\mathbb{Z}[\theta]) = -23$: <u>squarefree</u>, by prop, $\mathbb{Z}[\theta] = \mathcal{O}_K$

EXAMPLE

$K=\mathbb{Q}(\theta)$, $\theta^3-\theta-4=0$
$\operatorname{disc}(\mathbb{Z}[\theta]) = 2^2(-107)$
$\therefore [\mathcal{O}_K : \mathbb{Z}[\theta]] = 1$ or 2

<u>Case 1: $[\mathcal{O}_K : \mathbb{Z}[\theta]] = 2$</u>
$\mathbb{Z}[\theta] \subseteq \mathcal{O}_K \subseteq \frac{1}{2}(\mathbb{Z}[\theta])$

Check: $\{\frac{\varepsilon_0}{2} + \frac{\varepsilon_1}{2}\theta + \frac{\varepsilon_2}{2}\theta^2 \mid \varepsilon_i \in \{0,1\}\} =: S$
Does <u>S have integral elements</u>?

At most 1 elem of S is integral over $\mathbb{Z}$, excluding $0, \frac{\varepsilon_i}{2}\theta^i$?
Check the rest, for $s\in S$, $m_s: \mathbb{Q}(\theta) \longrightarrow \mathbb{Q}(\theta)$
$\alpha \longmapsto s\alpha$

M: matrix rep of m_s w.r.t. $\{\theta^i \mid 0\leq i\leq 2\}$

Define <u>$f_s(x)$</u>$:= \det(M - Ix)$, it has degree 3
$f_s(s)=0 \Rightarrow f_s$: irred $\forall s\notin\mathbb{Q}$ ($[K:\mathbb{Q}]=3 \Rightarrow [\mathbb{Q}(s):\mathbb{Q}] = 1$ or 3, but $s\notin\mathbb{Q} \Rightarrow [\mathbb{Q}(s):\mathbb{Q}]=3$)

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$\forall x \in S \setminus S_0$, $x \notin \mathbb{Q}$, thus $\mathbb{Q}(x) = k$

If $f_s \in \mathbb{Z}[x]$, $s \in \mathcal{O}_k$, then $\alpha := \frac{\theta}{2} + \frac{\theta^2}{2}$: integral $\Rightarrow \mathbb{Z}[\theta] + \mathbb{Z}\alpha = \mathcal{O}_k$

$\exists\, \beta \in \mathcal{O}_k$, s.t. $\mathcal{O}_k = \mathbb{Z}[\beta]$ (☆)

In our case, $\mathrm{disc}(\mathbb{Z}[\alpha]) = -107$, $\mathcal{O}_k = \mathbb{Z}[\alpha]$

DEDEKIND'S EXAMPLE

$K = \mathbb{Q}(\theta)$, $\theta^3 - \theta^2 - 2\theta - 8 = 0$

$\mathcal{O}_k$ cannot be generated by $\{\alpha^i\}$, $0 \le i \le 2$ (△)

Conceptual Proof / Evaluation

Assume $\exists\, \alpha \in \mathcal{O}_k$, s.t. $\mathcal{O}_k = \mathbb{Z}[\alpha] \cong \mathbb{Z}[x]/f_{\alpha,\mathbb{Q}}(x)$

$p \in \mathbb{Z}$: prime, $\mathbb{F}_p[x]/f_{\alpha,\mathbb{Q}}(x) \cong \mathcal{O}_k/p\mathcal{O}_k \longrightarrow \mathbb{F}_p$ has at most p ring homo $\mathcal{O}_k/p\mathcal{O}_k \to \mathbb{F}_p$

However, $2\mathcal{O}_k = \prod_{i=1}^{3} P_i$, P_i : distinct prime ideals of $\mathcal{O}_k$

$\mathcal{O}_k/2\mathcal{O}_k = \prod_{i=1}^{3} \mathcal{O}_k/P_i$, $\mathcal{O}_k/P_i \cong \mathbb{F}_2 \Rightarrow \mathcal{O}_k/2\mathcal{O}_k \to \mathbb{F}_2$ has 3 ring homo $\rightarrow\!\!\!\!\!\times$

Chinese remainder Thm

DEDEKIND DOMAINS AND DECOMPOSITION OF IDEALS

$\mathbb{Q} \hookleftarrow \mathbb{Z} \rightsquigarrow \alpha \in \mathbb{Z} \setminus \{0\}$, $\alpha = (\prod_{i=1}^{n} p_i^{e_i})(u)$, $u = \{\pm 1\}$, p_i : prime (PID $\Rightarrow$ UFD)

$\hookrightarrow \quad \hookrightarrow$

Dedekind domain $\Rightarrow$ decomposition of ideals (by integral domain)

$K \hookleftarrow \mathcal{O}_k$, $0 \neq I \trianglelefteq \mathcal{O}_k$, $\exists$ max ideal P_i s.t. $I = \prod_{i=1}^{n} P_i$ (P_i need not be distinct)

THEOREM

Given A: Dedekind domain, then $(0) \neq I \trianglelefteq A$, there are max ideals $P_i \trianglelefteq A$ s.t. $I = \prod_{i=1}^{n} P_i$ and this decomposition is unique up to rearrangement.

LEMMA

Ascending Chain Condition proof

Given I as in Thm above, then there are max ideals $P_i \trianglelefteq A$ s.t. $I \supseteq \prod_{i=1}^{n} P_i$ ← (△)